

AERO·TUNNEL: A Real-Time D2Q9 Lattice–Boltzmann Smoke Tunnel in the Browser

Numerical Methods Note

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Abstract

This note documents the mathematical model and numerical method behind AERO·TUNNEL, an interactive two-dimensional wind tunnel that runs entirely in a web browser (single-file HTML/JavaScript, no external solver). The flow is computed with a D2Q9 lattice–Boltzmann method (LBM) using the BGK single-relaxation-time collision operator, stabilised at higher Reynolds numbers by a Smagorinsky-type eddy-viscosity closure evaluated from the local non-equilibrium momentum flux. Aerodynamic forces are obtained by momentum exchange on boundary links, so the displayed lift and drag coefficients are *measured* from the simulation rather than taken from lookup data. This document states the discrete equations exactly as implemented, describes the boundary treatment, force evaluation, test-body geometry, tracer advection, and live diagnostics, and closes with an honest account of the scheme’s validity limits.

1 Governing model

The solver advances the discrete-velocity Boltzmann equation on the standard D2Q9 lattice with BGK relaxation [1, 2]:

$$f_i(\mathbf{x} + \mathbf{c}_i \Delta t, t + \Delta t) = f_i(\mathbf{x}, t) - \frac{1}{\tau} \left[f_i(\mathbf{x}, t) - f_i^{\text{eq}}(\rho, \mathbf{u}) \right], \quad i = 0, \dots, 8, \quad (1)$$

with lattice units $\Delta x = \Delta t = 1$. The nine discrete velocities are $\mathbf{c}_0 = (0, 0)$, the four axis vectors $(\pm 1, 0)$, $(0, \pm 1)$, and the four diagonals $(\pm 1, \pm 1)$, with weights

$$w_0 = \frac{4}{9}, \quad w_{1\dots 4} = \frac{1}{9}, \quad w_{5\dots 8} = \frac{1}{36}. \quad (2)$$

Macroscopic density and momentum are the zeroth and first moments,

$$\rho = \sum_i f_i, \quad \rho \mathbf{u} = \sum_i f_i \mathbf{c}_i, \quad (3)$$

and the equilibrium is the usual second-order Hermite truncation

$$f_i^{\text{eq}} = w_i \rho \left[1 + 3(\mathbf{c}_i \cdot \mathbf{u}) + \frac{9}{2}(\mathbf{c}_i \cdot \mathbf{u})^2 - \frac{3}{2} \mathbf{u} \cdot \mathbf{u} \right], \quad (4)$$

which is second-order accurate in Mach number about the lattice sound speed $c_s = 1/\sqrt{3}$. A Chapman–Enskog expansion of (1) recovers the weakly compressible Navier–Stokes equations with kinematic viscosity and pressure

$$\nu = c_s^2 \left(\tau - \frac{1}{2} \right) = \frac{1}{3} \left(\tau - \frac{1}{2} \right) \iff \tau = 3\nu + \frac{1}{2}, \quad p = c_s^2 \rho. \quad (5)$$

The implementation stores the nine populations in flat `Float32Array` buffers over a 260×110 grid and evaluates (4) in fully expanded scalar form for speed.

Because (4) truncates at $O(u^2)$, the model is valid only well below c_s . The inflow speed is therefore capped at $u_0 \leq 0.15$ lattice units ($\text{Ma} = u_0/c_s \approx 0.26$); the user-facing speed slider spans $u_0 \in [0.020, 0.150]$.

2 Eddy-viscosity stabilisation (Smagorinsky-type)

Plain BGK becomes unstable when $\tau \rightarrow \frac{1}{2}$ (small ν , i.e. high Re) on a coarse grid. Following the LES-inspired construction of Hou et al. [4], the code augments the molecular relaxation time with a local eddy contribution computed from the non-equilibrium momentum flux, which in LBM is available algebraically, without finite differences:

$$\Pi_{\alpha\beta}^{(1)} = \sum_i c_{i\alpha} c_{i\beta} (f_i - f_i^{\text{eq}}), \quad \bar{Q} = \left(\Pi_{xx}^{(1)} \Pi_{xx}^{(1)} + \Pi_{yy}^{(1)} \Pi_{yy}^{(1)} + 2 \Pi_{xy}^{(1)} \Pi_{xy}^{(1)} \right)^{1/2}. \quad (6)$$

The effective relaxation time used in the collision step is the positive root of the consistency relation between $\bar{Q}$ and the Smagorinsky eddy viscosity $\nu_t = (C_s \Delta)^2 |S|$ (with filter width $\Delta = \Delta x = 1$):

$$\tau_{\text{eff}}(\mathbf{x}) = \frac{1}{2} \left(\tau_0(\mathbf{x}) + \sqrt{\tau_0(\mathbf{x})^2 + \frac{18 C_s^2 \bar{Q}}{\rho}} \right), \quad C_s^2 = 0.03, \quad (7)$$

where $\tau_0 = 3\nu + \frac{1}{2} + \sigma(x)$ includes the outlet sponge described in §3. Equation (7) is evaluated pointwise every step; collisions then use $\omega = 1/\tau_{\text{eff}}$ in (1). It must be stressed that this is used here as a *stabilisation device* that adds dissipation only where the non-equilibrium stress is large (shear layers, wakes); no claim of validation-grade LES is made – there is no wall model, no dynamic procedure, and the grid is far from LES resolution requirements.

3 Streaming, boundaries, and initialisation

Streaming. Populations are propagated with in-place directional sweeps (pull ordering chosen per direction so that no temporary buffer is required). Screen coordinates place y increasing downward; the direction labels are bookkeeping only.

Inlet and far field. The left column is a velocity inlet: populations are reset each step to $f_i^{\text{eq}}(\rho = 1, \mathbf{u} = (u_0, 0))$. The top and bottom rows are treated the same way, which acts as a far-field Dirichlet condition. This is deliberate and cheap, but it makes the domain a *closed test section*: at large incidence the model blocks an appreciable fraction of the tunnel height and wall interference inflates the measured coefficients (see §11).

Outlet. The right boundary uses a zero-gradient (convective-type) copy, $f_i(x_{\text{max}}, y) \leftarrow f_i(x_{\text{max}} - 1, y)$, preceded by a *viscous sponge*: over the last $W_s = 28$ columns the base relaxation time is increased by

$$\sigma(x) = 0.6 \left(\frac{x - (x_{\text{max}} - W_s)}{W_s} \right)^2, \quad x > x_{\text{max}} - W_s, \quad (8)$$

so that acoustic and vortical disturbances are damped before they can reflect from the truncated boundary. The combination behaves as a simple open-outlet approximation.

Solid bodies. The body is rasterised onto the lattice as a solid mask. On the fluid–solid interface the code applies full bounce-back: for every boundary cell, each population that arrived from a fluid neighbour during streaming is reflected into its opposite direction and deposited back into that fluid neighbour. Only *surface* cells participate; the solid interior never exchanges populations with the flow. Bounce-back places the effective wall approximately halfway between lattice nodes and is first-order accurate on the staircase geometry used here.

Initialisation. The field is initialised at full freestream rather than ramped from rest – an empirically motivated choice: a global ramp was observed to converge to a spurious low-throughput branch in this closed section. To avoid the acoustic ring of an impulsive start against the body, the initial velocity is faded to zero near the surface using a two-pass chamfer distance transform $d(\mathbf{x})$ to the solid and a smoothstep profile,

$$\mathbf{u}(\mathbf{x}, 0) = u_0 s^2 (3 - 2s) \hat{\mathbf{e}}_x, \quad s = \min(1, d(\mathbf{x})/10). \quad (9)$$

A watchdog monitors a density probe one chord downstream of the body; if ρ leaves $(0.2, 5]$ or becomes non-finite anywhere in the collision loop, the offending cell (or the whole field) is re-equilibrated and the user is notified.

4 Aerodynamic forces: momentum exchange

The hydrodynamic force on the body is evaluated with the momentum-exchange method of Ladd [6]: every boundary link that connects a surface cell to a fluid cell transfers momentum $2f_i \mathbf{c}_i$ per bounce-back event. Summing over all cut links,

$$\mathbf{F} = \sum_{\text{links } (\mathbf{x}, i)} 2 f_i(\mathbf{x}) \mathbf{c}_i, \quad (10)$$

where the sum counts *only populations that arrived from fluid neighbours* – links between two solid cells contribute nothing, which is the correct discrete statement of surface momentum flux. The force is accumulated over the n solver sub-steps of each rendered frame and converted to coefficients per unit span with $\rho_0 = 1$:

$$C_L = \frac{-\langle F_y \rangle}{\frac{1}{2} u_0^2 L}, \quad C_D = \frac{\langle F_x \rangle}{\frac{1}{2} u_0^2 L}, \quad (11)$$

with reference length $L = c$ (chord, 83 cells) for wings and $L = D$ (diameter, 24.2 cells) for the cylinder. The sign of C_L accounts for the downward-positive screen y -axis. The HUD applies an exponential moving average, $\bar{C} \leftarrow 0.95 \bar{C} + 0.05 C$, while the time-history chart plots the raw per-frame samples so that vortex shedding remains visible as oscillation.

5 Test-body geometry

Wings use the standard NACA four-digit parametrisation [7]. With maximum camber m , camber position p , and thickness ratio t (all as chord fractions), the camber line and half-thickness are

$$y_c(x) = \begin{cases} \frac{m}{p^2} (2px - x^2), & 0 \leq x < p, \\ \frac{m}{(1-p)^2} [(1-2p) + 2px - x^2], & p \leq x \leq 1, \end{cases} \quad (12)$$

$$y_t(x) = 5t (0.2969\sqrt{x} - 0.1260x - 0.3516x^2 + 0.2843x^3 - 0.1036x^4), \quad (13)$$

where the final coefficient -0.1036 (rather than the classical -0.1015) closes the trailing edge. Two implementation simplifications are worth noting for the reviewer: (i) thickness is applied perpendicular to the *chord* rather than to the camber line – a standard small-camber approximation, exact for symmetric sections; (ii) y_t is floored at $0.012c$ so that the rasterised section never thins below one lattice cell. The section is pitched about the 40%-chord point; a point (x, y) of the lattice is tested by rotating it into the airfoil frame by $-\alpha$ and checking $y_c - y_t \leq y_n \leq y_c + y_t$. The flat plate is generated as a degenerate section with $t = 0.024$; the cylinder has radius $r = 0.11 N_y$. When the incidence slider moves, cells freed by the rotating body are re-equilibrated from the average of their surviving fluid neighbours rather than from freestream, which suppresses the acoustic pulse that a freestream fill would emit on every slider tick.

6 Smoke tracers

Between 200 and 6000 passive tracers (user-adjustable) are advected with a second-order Runge–Kutta midpoint scheme on the bilinearly interpolated velocity field,

$$\mathbf{x}^{n+1} = \mathbf{x}^n + \mathbf{u}\left(\mathbf{x}^n + \frac{1}{2}\mathbf{u}(\mathbf{x}^n)\right), \quad (14)$$

with $\Delta t = 1$ per solver sub-step. The midpoint evaluation keeps streaklines attached to curved streamlines where forward Euler visibly drifts. Tracers that exit the domain or penetrate the solid mask are reseeded on a 26-lane rake at the inlet. Rendering draws the per-step displacement segment of each tracer into a persistent canvas that is exponentially faded (10% destination-out per frame), producing continuous streaklines rather than points; the fade is suspended when the simulation is paused so the pattern freezes rather than evaporates.

7 Live diagnostics

Reynolds number. Reported as $Re = u_0 L / \nu$ with the effective (possibly floored) viscosity; with the available ranges this spans roughly $Re \sim 30$ to 3×10^3 – model scale, not flight scale.

Strouhal number. For the cylinder, shedding frequency is estimated from upward zero-crossings of $C_L(t)$ about its mean over the recent history. Because the number of solver sub-steps per rendered frame adapts to machine speed, elapsed time between crossings is measured in *accumulated lattice steps*, making

$$St = \frac{f D}{u_0} \quad (15)$$

fully dimensionless and machine-independent. The estimate is gated on a minimum sample count and a minimum lift oscillation amplitude, and is suppressed for wings where it is not meaningful.

Flow-regime classifier. A HUD badge classifies the instantaneous regime by combining incidence with the *measured* unsteadiness ΔC_L (peak-to-peak lift over the last ~ 120 samples, lightly low-pass filtered):

Label	Condition	Interpretation
STABLE FLOW	$ \alpha < 8^\circ$ and $\Delta C_L \leq 0.18$	attached, steady lift
PRE-STALL	$ \alpha \geq 8^\circ$ or $\Delta C_L > 0.18$	incipient separation
STALLING	$ \alpha \geq 12^\circ$ or $\Delta C_L > 0.35$	unsteady, separating
POST-STALL	$ \alpha \geq 18^\circ$	fully separated wake

The angle criteria respond instantly to the control; the fluctuation criterion lets measured unsteadiness promote the state even at moderate incidence.

8 Stability envelope

Rather than guessing a safe viscosity, the admissible lower bound $\nu_{\min}(u_0, \alpha)$ was *measured* on this exact solver: a NACA 0012 sweep over speed and incidence recorded the smallest ν that survives 8000 steps at each operating point. The runtime floor bilinearly interpolates this table over $(u_0, |\alpha|)$, applies a $1.5\times$ margin to the excess above the baseline $\nu = 0.004$, and an additional $1.2\times$ factor for bluff bodies (cylinder, plate), which are mapped to an equivalent incidence of 30° . When the floor is active the viscosity readout is starred (ν^*) so the user knows the requested value was raised. The physical reason the floor grows so steeply with $|\alpha|$ is blockage: near $\alpha = 45^\circ$ the model obstructs over half the section height and local speeds far exceed u_0 , pushing the local Mach and grid Reynolds numbers beyond what the scheme tolerates.

9 Rendering of derived fields

For completeness (these operations never feed back into the solver): vorticity is computed by centred differences, $\omega = \partial_x u_y - \partial_y u_x$; the pressure coefficient follows from the LBM equation of state, $C_p = ((\rho - 1)/3)/(\frac{1}{2}u_0^2)$. Before upsampling to screen resolution, scalar values inside the solid are replaced by the average of fluid neighbours (so bilinear interpolation does not smear a dark halo along the rasterised staircase), and the field is smoothed with two rounds of a separable $\frac{1}{4}(1, 2, 1)$ binomial filter, whose frequency response is exactly zero at the grid checkerboard frequency – eliminating the odd–even noise characteristic of LBM without touching the solved fields. The analytic body outline is drawn on a separate canvas layer above the field.

10 Numerical parameters

Quantity	Symbol	Value	Notes
Grid	$N_x \times N_y$	260×110	lattice units, $\Delta x = \Delta t = 1$
Chord	c	83 cells	$0.32 N_x$; pivot at $0.40 c$
Cylinder diameter	D	24.2 cells	$r = 0.11 N_y$
Inflow speed	u_0	0.020–0.150	$\text{Ma} \leq 0.26$
Molecular viscosity	ν	0.004–0.050	floored by stability envelope
Relaxation time	τ	$3\nu + \frac{1}{2}$	Eq. (5)
Smagorinsky constant	C_s^2	0.03	Eq. (7)
Sponge	$W_s, \sigma_{\max}$	28 cells, 0.6	quadratic ramp on τ
Tracers	N_p	200–6000	RK2 midpoint advection
Sub-steps per frame	n	4–14	adaptive to frame time
Force smoothing	–	EMA, $\lambda = 0.05$	HUD only; charts use raw samples

11 Validity and limitations

The reader should treat the displayed coefficients as *qualitative*. Specifically: (i) the flow is two-dimensional, so three-dimensional relief effects, tip vortices, and spanwise instabilities are absent; (ii) Reynolds numbers are model-scale (10^1 – 10^3), far below flight values, so boundary layers are comparatively thick and stall occurs earlier and more gently than at full scale; (iii) the closed test section with $c/N_y \approx 0.75$ produces significant wall interference and blockage at high incidence – no blockage correction is applied; (iv) the staircase bounce-back wall is first-order accurate, and at 83 cells per chord the boundary layer is under-resolved, with the eddy-viscosity term absorbing what the grid cannot represent; (v) the scheme is weakly compressible, and spurious acoustics are controlled rather than eliminated (sponge, local fades, watchdog). What survives these caveats – and what the tool is built to show – are the *trends*: the linear lift slope and its magnitude ($dC_L/d\alpha$ below stall), camber shifting the zero-lift angle, stall onset and hysteresis-free post-stall lift collapse, the drag polar shape, bluff-body separation, and Kármán shedding with a Strouhal number near the expected $St \approx 0.2$ range for the cylinder at these Reynolds numbers.

References

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